Total No. of Questions : 8]

PC-2675

SEAT No. :

[Total No. of Pages : 2

[6354]-841

B.E. (Computer Engineering) (Honours)

Machine learning and Data Science

(2019 Pattern) (Semester - VII) (410501)

Time : 2½ Hours] *[Max. Marks : 70*

Instructions to the candidates:

1) ***Solve Q.1 or Q.2, Q.3 or Q.4, Q.5 or Q.6,Q.7 or Q.8.***
2) ***Neat diagrams must be drawn whenever necessary.***
3) ***Figures to the right indicate full marks.***

Q1) a) Explain the role of dendrograms in Choosing number clusters in Hierarchical clustering? **[6]**

b) Explain supervised and unsupervised learning in detail **[6]**

c) What is the K-Mean algorithm? Explain its steps with an example. **[6]**

OR

Q2) a) What is Density-Based Spatial Clustering? Explain with suitable examples. **[6]**

b) For what type of data, Density-Based Spatial Clustering is suitable? Which parameters are required by DBSCAN algorithm? **[6]**

c) Explain KNN algorithm with an example. **[6]**

Q3) a) Explain artificial neural networks? What are the types of artificial neural networks? **[6]**

b) What is Perceptron? Explain the process of training a perceptron? **[6]**

c) Explain Generalized delta learning rule in detail. **[6]**

P.T.O.

OR

Q4) a) Explain Feed-forward and feedback neural networks. **[6]**

b) Explain the term Multilayer perceptron's with its limitations **[6]**

c) Explain Back propagation Algorithm in detail **[6]**

Q5) a) Explain CNN algorithm. Enlist and explain any two types of CNN models. **[9]**

b) Explain the terms "Valid Padding" and "Same Padding" in CNN. List down the hyper parameters of a Pooling Layer. **[8]**

OR

Q6) a) Explain in detail gradient descent optimization? **[9]**

b) Difference between Recursive Neural Network and Recurrent Neural Network **[8]**

Q7) a) Explain the following terms: **[9]**

a) Stemming

b) Lemmatization

b) Describe tokenization with an example **[8]**

OR

Q8) a) What is preprocessing? Explain method of text preprocessing **[9]**

b) Explain topic modeling with Latent Dirichlet Allocation algorithm **[8]**

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